LLM-aided Human-Robot Interfacing for A Prompt-to-Control Application

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Introduction

This work presents a novel solution to interface and control a mobile robot with human operators leveraging a custom large language model (LLM) hosted on a server. Here, operators can provide commands as freeform text to the server via a user interface without a need of any prior knowledge on the machine language. In the server, a LLM that has been customized with the knowledge of the environment is used to break down the directives into a high-level plan in terms of

i) Go where?
ii) Find what? and
iii) Do what?,

which are then shared with the robot. Upon receiving the plan, the robot autonomously navigates to the designated location and identifies target objects utilizing an optimized tiny artificial intelligent (AI) model to ensure high performance in real-time. The demo validates the effectiveness of generating accurate plans even for the broader commands within the context and the ability to perform the task in real-time.

Architecture & Hardware

- **Robot:** Waveshare "JetBot ROS AI Kit", including a Nvidia Jetson Nano developer module with following sensors [3].
  - RPLiDAR A1
  - Inertial measurement unit (IMU) sensor
  - Motor encoder sensors
  - Intel RealSense D415 Camera
- **Server Computer:** Uses Ubuntu 20.04 and ROS with NVIDIA Ge GPU.
  - LLM: Mistral 7B Instruct model [2]
- **Demo platform:** Enclosed area reflecting a downscaled layout of a workplace.
  - Includes a sets of destinations $\mathcal{D}$ = {coffee room, meeting room, professor's office, robotics lab}
  - Objects $\mathcal{O}$ = {bin, coffee machine, professor, robot, TV}

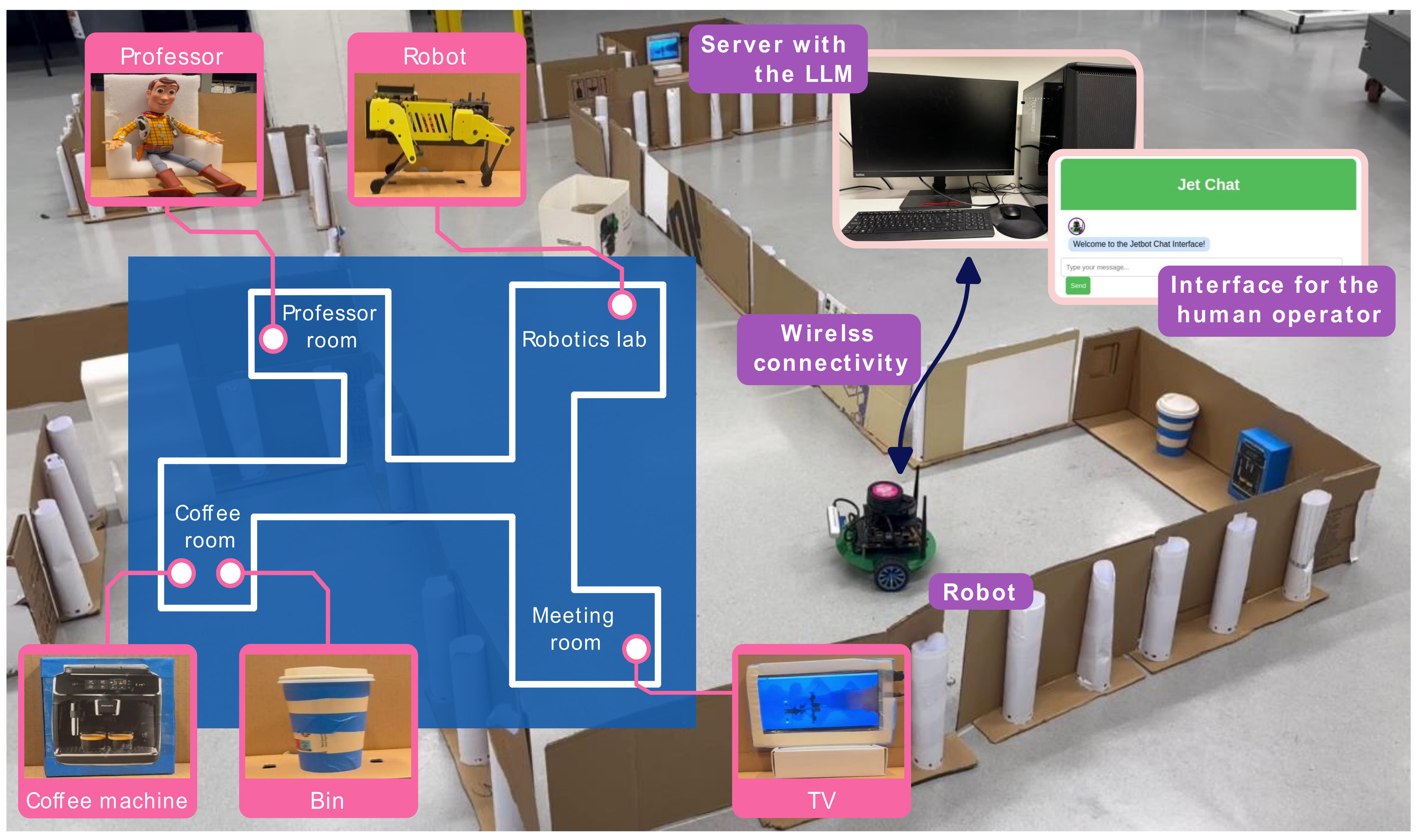

Fig. 1: The robotic platform highlighting the layout and the association of the objects in $\mathcal{O}$ with the destinations in $\mathcal{D}$.

System Implementation

Step 1 – High-Level Planning:

- **User Input:** Freeform natural language prompts via a chat-box interface.
- **LLM Interpretation:** The directives are structured as a high-level plan [5]
  - Go where? *Identify the destination.*
  - Find what? *Determine the target object.*
  - Do what? *Define the action to be performed.*
- **Output:** A structured tuple (destination: D, object: O, action: A) referred to as the DOA message.

TABLE I: Exemplary prompts and corresponding LLM outputs.

User prompt	Output
Take this file to the professor at the professor's room.	D: professor's room O: professor A: take this file to the professor
Check whether the video feed works in the meeting room.	D: meeting room O: TV A: check whether the video feed works
Switch on the robot.	D: robotics lab O: robot A: switch on
Please make me a coffee.	D: coffee room O: coffee machine A: make coffee

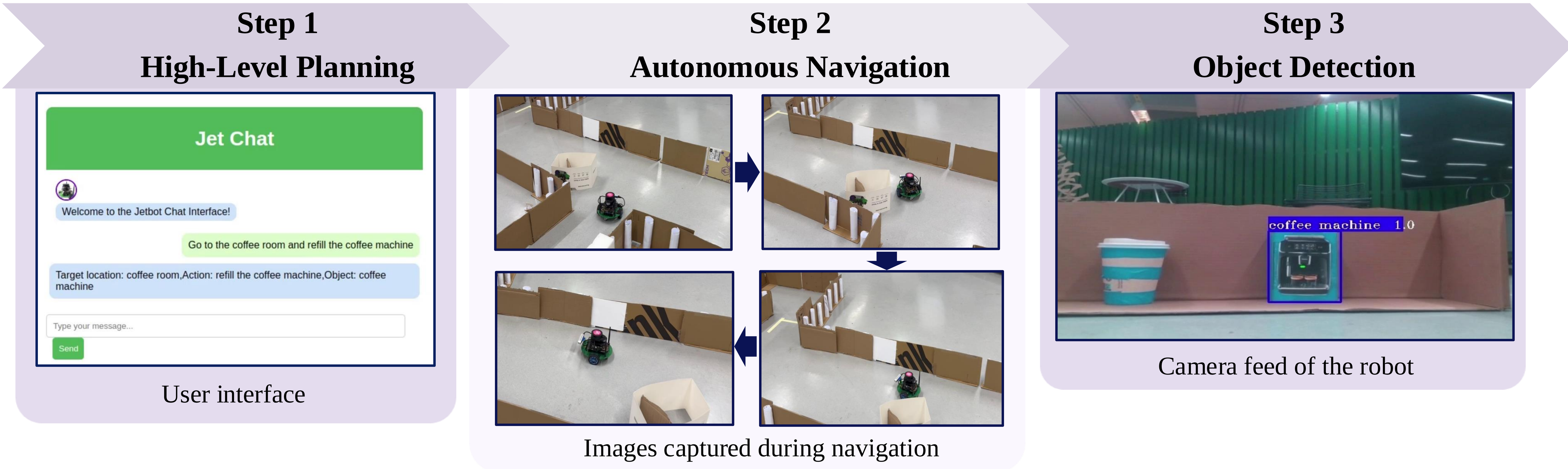

Fig. 2: Flow of the system implementation highlighting the three key steps along the corresponding outputs from an actual demo.

Step 2 – Autonomous Navigation:

- **Input:** The DOA message.
- **Operation:** Extracts the destination D and autonomously move towards it.
  - Uses the TurtleBot3 navigation package [6].
  - Employs sensor data for real-time obstacle detection and dynamic path planning.
- **Output:** Motor control commands.

Step 3 – Object Detection:

- **Input:** Onboard camera feed at the destination.
- **Operation:** Detects object O using a YOLOv4-tiny model.
  - The model is trained on a custom dataset with the various orientations of objects in $\mathcal{O}$.
  - The model is optimized with TensorRT for efficient real-time inference [4].
- **Output:** Flag indicating whether the object is detected or not.

Conclusions

This work demonstrates an effective approach to human-robot interfacing using LLMs for real-time control. The solution bridges the communication gap, making robot control more accessible.

Future directions

- Expand LLM capabilities for multi-task planning.
- Multi-robot collaboration in dynamic environments.

References

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